

Quantum of Action as Two Conjugate Cells Resolves Caustics and Wigner Negativity

Brian S. Mulloy^{✉*}
(Dated: 2026-05-04)

A quantum state has finite action in phase space. Heisenberg's uncertainty combined with de Gosson's topology structures Planck's quantum of action h as two conjugate cells of size $h/2$, one squeezed along position, the other along momentum. Caustics in the WKB density and negativity in the Wigner function are mathematical artifacts because both formalisms calculate below this scale. At the physical scale, the state is everywhere finite and non-negative.

A quantum state has finite action in phase space. The Heisenberg and Robertson–Schrödinger uncertainty principles [5, 6],

$$\sigma_q \sigma_p \geq h/2, \quad (1)$$

forbid the state from being a point: zero area is incompatible with finite uncertainty.

De Gosson identified the state's extent in phase space as the Fermi blob [?], the elliptical region defined by its covariance with semi-axes $\Delta q = \sqrt{2}\sigma_q$ and $\Delta p = \sqrt{2}\sigma_p$. Its covariance area,

$$A = \pi \Delta q \Delta p \geq h/2, \quad (2)$$

is distinct from the system's dynamical action, saturating at $h/2$ only for the exact harmonic ground state. A Fermi blob of covariance area $A > h/2$ is not featureless. Its interior holds two conjugate quantum blobs [11, 12],

$$a_q = \pi \delta q \Delta p = h/2, \quad a_p = \pi \Delta q \delta p = h/2, \quad (3)$$

each carrying half the Planck quantum of action and together summing to h (Fig. 1). The narrow widths

$$\delta q = \hbar/\Delta p, \quad \delta p = \hbar/\Delta q \quad (4)$$

are reciprocal to the Fermi-blob semi-axes, fixed by the same Heisenberg constant in Eq. (??). Equation (??) is itself an uncertainty relation between conjugate resolutions inside a state of fixed covariance. As A grows, Δp and Δq grow while δq and δp shrink: Heisenberg binds the state at both scales. Gromov's non-squeezing theorem [8] forbids any canonical transformation from compressing a quantum blob below $h/2$. Planck's quantum of action is the floor of phase space.

Two pathologies arise when formalisms breach this floor. The WKB density [1]

$$|\psi_{\text{WKB}}(q)|^2 = \frac{N^2}{p(q)} \cos^2(S(q)/\hbar - \pi/4) \quad (5)$$

diverges at every classical turning point. The Wigner–Weyl distribution [3]

$$W(q, p) = \frac{1}{\pi \hbar} \int_{-\infty}^{\infty} \psi^*(q+y) \psi(q-y) e^{2ipy/\hbar} dy \quad (6)$$

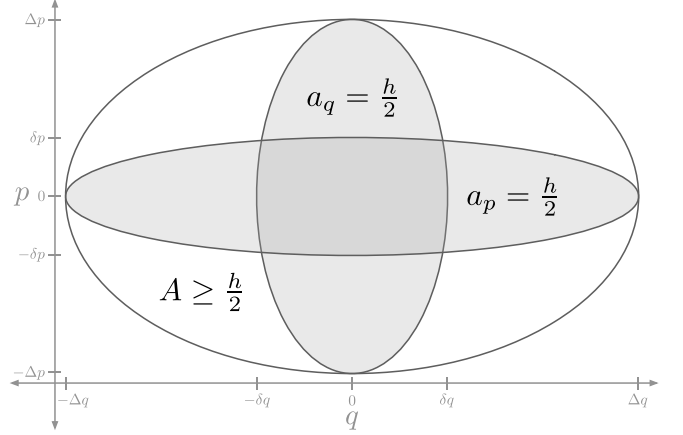


FIG. 1. **Conjugate structure of the quantum state** (schematic, 1D case). The Heisenberg limit $\sigma_q \sigma_p \geq h/2$ admits semi-axes $\Delta q = \sqrt{2}\sigma_q$, $\Delta p = \sqrt{2}\sigma_p$; the state's 1σ contour is its Fermi blob [?], of area $A = \pi \Delta q \Delta p \geq h/2$. Inside sit two conjugate quantum blobs [11, 12], $a_q = \pi \delta q \Delta p = h/2$ squeezed along position and $a_p = \pi \Delta q \delta p = h/2$ squeezed along momentum, summing to $h = a_q + a_p$. Their narrow widths are $\delta q = \hbar/\Delta p$ and $\delta p = \hbar/\Delta q$. As A grows, Δp and Δq grow while δq and δp shrink: Heisenberg holds at every scale shown.

takes negative values for any state beyond the ground state. Connection formulas and Airy uniformization [2] patch the caustic locally; the Husimi Q -function [4] smooths Wigner with a fixed coherent-state kernel of unit width and erases every scale below it. Each remedy treats one symptom in one formalism.

The two pathologies share a single root. WKB evaluates density at a precise q ; Wigner evaluates it at an exact (q, p) . Each assumes a zero-area point evaluation ($\delta q \rightarrow 0$ or $\delta p \rightarrow 0$), which is physically forbidden [?]. By Eq. (??) this demands an infinite conjugate spread ($\Delta p \rightarrow \infty$ or $\Delta q \rightarrow \infty$) and unbounded covariance area ($A \rightarrow \infty$). Caustics and Wigner negativity are mathematical artifacts of this forbidden limit, not features of the state.

The two conjugate quantum blobs are the physical scale of the state. Because pointwise evaluation ignores this floor, we adapt the phase-space filtering of [?] with a kernel set by the state's own quantum blobs. The conjugate quantum blobs of Fig. 1, as Gaussian filters of

phase-space area $h/2$, give

$$G_{\delta q}(q, p) = \frac{1}{\pi} \exp\left(-\frac{q^2}{\delta q^2} - \frac{p^2}{\Delta p^2}\right), \quad (7)$$

$$G_{\delta p}(q, p) = \frac{1}{\pi} \exp\left(-\frac{q^2}{\Delta q^2} - \frac{p^2}{\delta p^2}\right), \quad (8)$$

$G_{\delta q}$ squeezed along position and $G_{\delta p}$ along momentum. Each kernel is itself a quantum blob at the $h/2$ floor. Unlike Husimi's fixed-width kernel, $G_{\delta q}$ tracks the state's own covariance.

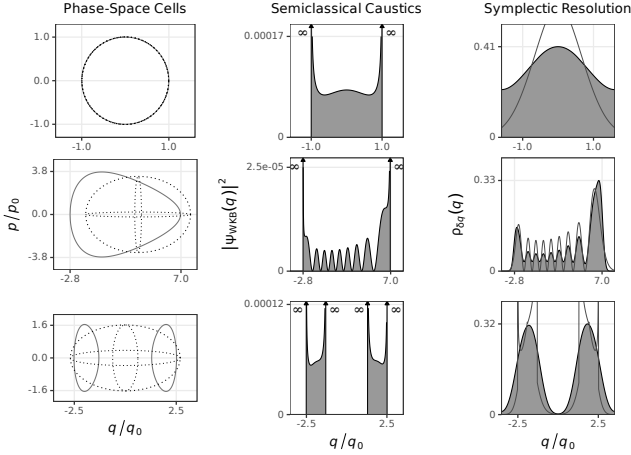


FIG. 2. **Semiclassical resolution.** Rows: harmonic ground state, Morse $n = 8$, symmetric double-well sub-barrier doublet at $A_{BS}/A_0 \approx 1.976$. *Left:* Classical orbit at E_n with the conjugate quantum blobs overlaid. *Center:* WKB density with caustic divergences (rendered as infinity arrows) at every classical turning point. *Right:* Resolved density $\rho_{\delta q}(q)$ with the per-orbit Airy uniform density [2] overlaid. The Morse row preserves the $n + 1$ WKB lobes; divergences become finite peaks. Per-well Airy lobe maxima clip the panel at every inner turning point of the double-well; $\rho_{\delta q}$ is everywhere finite.

For the classical microcanonical density $W_{cl}(q, p) \propto \delta(H - E)$ at the Bohr–Sommerfeld energy E_n , the orbit's time-averaged second moments give $\Delta q = \sqrt{2\langle(q - \langle q \rangle)^2\rangle_{orbit}}$ and $\Delta p = \sqrt{2\langle p^2 \rangle_{orbit}}$. The marginal of the convolution over p is

$$\rho_{\delta q}(q) = \int (W_{cl} * G_{\delta q})(q, p) dp, \quad (9)$$

finite at the turning points where $|\psi_{WKB}|^2$ diverges, and strictly positive everywhere due to the kernel's Gaussian tails. Figure ?? shows three systems. The harmonic ground state collapses to a single Gaussian. The Morse oscillator at $n = 8$ preserves the $n + 1$ Bohr–Sommerfeld interference lobes; the WKB caustics become finite peaks. For the symmetric double-well sub-barrier doublet, per-well Langer–Airy constructions place

artificial maxima at the inner turning points; $\rho_{\delta q}(q)$ is everywhere finite and seamlessly resolves the physical barrier penetration.

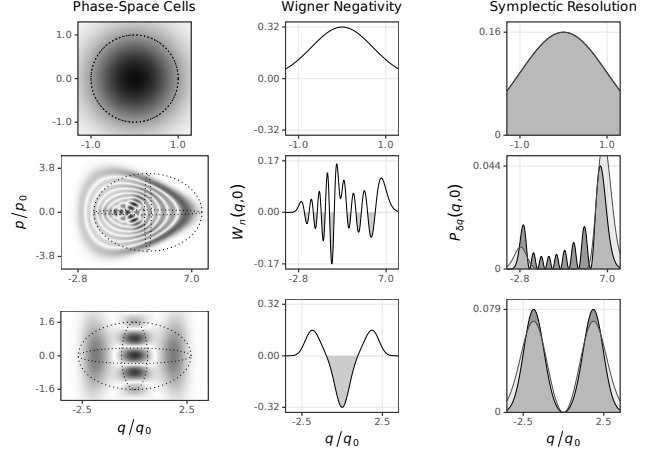


FIG. 3. **Wigner resolution.** Same three systems as Fig. ??, evaluated as Schrödinger eigenstates: harmonic $n = 0$, Morse $n = 8$, double-well $n = 1$ (antisymmetric lower-doublet partner with exact node $\psi_1(0) = 0$). Kernel widths from the Robertson–Schrödinger covariance of ψ_n . *Left:* Wigner distribution $W_n(q, p)$. *Center:* Cross-section $W_n(q, 0)$. The double-well row reaches $W_1(0, 0) = -0.32$; the wavefunction's exact node lies at the deepest Wigner negativity. *Right:* $P_{\delta q}(q, 0)$ with Husimi $Q(q, 0)$ overlaid. Husimi's fixed width $q_0 = 1$ exceeds δq for the Morse row and smooths the multi-lobe structure into two broad peaks at the turning points; δq tracks the wavefunction. At the double-well node $P_{\delta q}$ is strictly positive (numerically 5×10^{-6} relative to peak ~ 0.08).

The same kernel applies in the wave formalism, with $\Delta q = \sqrt{2\langle(q - \langle q \rangle)^2\rangle_{\psi_n}}$ and $\Delta p = \sqrt{2\langle p^2 \rangle_{\psi_n}}$ from the eigenstate's Robertson–Schrödinger covariance. The convolved Wigner density

$$P_{\delta q}(q, p) = (W * G_{\delta q})(q, p) \quad (10)$$

is non-negative everywhere: convolution with a quantum-blob kernel sends any Wigner function to a non-negative one [9?]. Figure ?? shows the three systems as eigenstates. The Morse row at $n = 8$ has $n + 1$ alternating-sign Wigner lobes; $P_{\delta q}$ preserves the lobe envelope and is non-negative throughout, while Husimi smooths away the interior interference, leaving only two broad peaks at the classical turning points. The double-well antisymmetric partner ψ_1 has an exact wavefunction node at $q = 0$, which coincides with $W_1(0, 0) = -0.32$, the deepest point of the Wigner negativity — the same unphysical point-evaluation failing in two different formalisms. At resolution δq , $P_{\delta q}(q, 0)$ is strictly positive throughout, including across the node and through the barrier where the Gaussian tails extend from each well.

A cat state has no classical orbit and no Bohr–Sommerfeld action. The Fermi blob is still finite, set

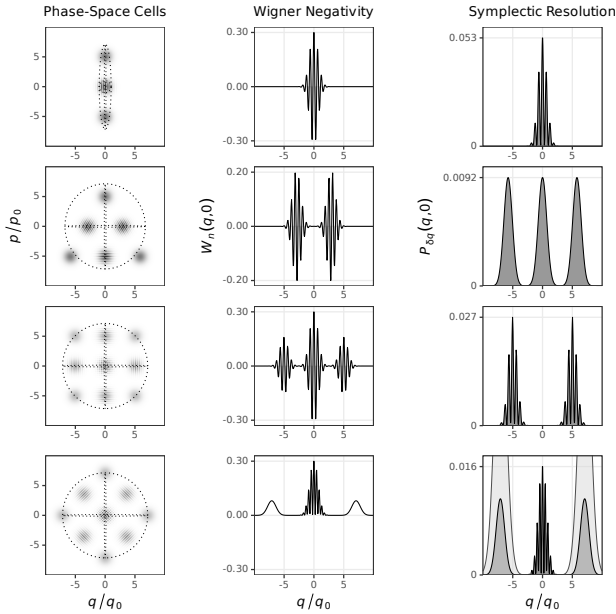


FIG. 4. **Cat states.** Coherent superpositions of n Gaussian wavepackets with no classical orbit and no Bohr–Sommerfeld condition. Kernel widths from the Robertson–Schrödinger covariance of ψ directly. Rows: 2-cat at $(0, \pm 5)$; 3-cat at the equilateral triangle apex $(0, 5)$ and base $(\pm 10/\sqrt{3}, -5)$; 4-cat compass at $(\pm 5, 0)$, $(0, \pm 5)$ [7]; 4-cat compass rotated 45° to $(\pm 5, \pm 5)$. Columns as in Fig. ?? . Each row’s Wigner cross-section dips below zero — including the sub-Planck chessboard [7] of the rotated compass. $P_{\delta q}(q, 0)$ is non-negative for every row.

by the wavefunction’s covariance directly, with $\Delta q = \sqrt{2\langle (q - \langle q \rangle)^2 \rangle_\psi}$ and $\Delta p = \sqrt{2\langle p^2 \rangle_\psi}$. Figure ?? shows four cat configurations. Each row’s Wigner cross-section dips below zero — the central interference fringe of the 2-cat, the pair-fringe envelopes of the 3-cat, the chessboard of the axis-aligned compass, and the sub-Planck chessboard of the rotated compass. Zurek’s sub-Planck chessboard [7] is the conjugate quantum-blob width $\delta q = \hbar/\Delta p$. At resolution δq , every cross-section is non-negative.

A quantum state has finite action in phase space and

cannot be reduced to a point. Caustics and Wigner negativity are mathematical artifacts of ignoring this physical reality. That action is partitioned by Heisenberg’s uncertainty and de Gosson’s topology into a Fermi blob and two conjugate quantum blobs. At the physical scale, the state is everywhere finite and non-negative. All nu-

merical data, figure-generating code, and reproducibility scaffolding for this paper are available at <https://doi.org/10.17605/OSF.IO/EV8H6>.

* bmulloy@umich.edu

- [1] M. V. Berry, Semi-classical mechanics in phase space: A study of Wigner’s approach, *Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences* **287**, 237 (1977).
- [2] R. E. Langer, On the connection formulas and the solutions of the wave equation, *Physical Review* **51**, 669 (1937).
- [3] E. Wigner, On the quantum correction for thermodynamic equilibrium, *Physical Review* **40**, 749 (1932).
- [4] K. Husimi, Some formal properties of the density matrix, *Proceedings of the Physico-Mathematical Society of Japan. 3rd Series* **22**, 264 (1940).
- [5] H. P. Robertson, The uncertainty principle, *Physical Review* **34**, 163 (1929).
- [6] E. Schrödinger, Zum Heisenbergschen Unschärfeprinzip, *Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse* **14**, 296 (1930).
- [7] W. H. Zurek, Sub-Planck structure in phase space and its relevance for quantum decoherence, *Nature* **412**, 712 (2001).
- [8] M. Gromov, Pseudo holomorphic curves in symplectic manifolds, *Inventiones mathematicae* **82**, 307 (1985).
- [9] R. L. Hudson, When is the Wigner quasi-probability density non-negative?, *Reports on Mathematical Physics* **6**, 249 (1974).
- [11] M. A. de Gosson, Phase space quantization and the uncertainty principle, *Physics Letters A* **373**, 3896 (2009).
- [12] M. de Gosson and C. de Gosson, Symplectic polar duality, quantum blobs, and generalized Gaussians, *arXiv preprint* (2022), arXiv:2206.06334 [quant-ph].